

Hippocampal Synaptic Markers and Cognitive Recovery in Spinal Cord Injury: The Therapeutic Potential of Neural Stem Cell-Laden Carbon Nanotube-Based Fiber Scaffolds with Liposomal Hesperidin

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1. Methods

1.1. Structural Analysis of Scaffolds

The structural characteristics of the scaffolds were assessed using a field emission scanning electron microscope (FESEM, Tescan Mira3 LMU, Czech Republic) at an accelerating voltage of 10 kV. Prior to imaging, samples were affixed to metal stubs with conductive double-sided tape and coated with a thin layer of gold. Fiber diameters were quantified from FESEM images using ImageJ software, measuring at least 50 fibers for accuracy.

The morphology of NSCs and Hsd@lip cultured on the scaffolds was analyzed after a 2-day incubation. Samples underwent fixation in 2.5% glutaraldehyde at 4 °C for 3 hours, followed by rinsing with PBS and dehydration through a series of ethanol gradients (30%, 70%, 90%, 96%, and 100% v/v) for 10 minutes each. Finally, samples were placed in a desiccator, gold-coated, and examined via FESEM.

1.2. Hind Limb Motor Behavior

The day prior to the injury, the locomotion of all animals was evaluated using the 21-point Basso, Beattie, and Bresnahan (BBB) Locomotor Scale. Following the spinal cord injury in the rats, hind limb motor functions were re-assessed using the same BBB Scale. A score of 0 indicates no movement in the hind limbs, while a score of 21 signifies no motor deficits. One day after surgery, the rats were observed for 4 minutes while moving freely in a 1-m diameter arena, and their scores were recorded weekly for 4 weeks by two observers who were blinded to the group assignments.

1.3. Biochemical evolution

1.3.1. Measurement of Malondialdehyde (MDA) Levels

To assess the MDA levels, 15 g of TCA (Sigma-Aldrich) and 0.375 g of TBA (Merck, Germany) were first weighed. These compounds were then combined with 2 ml of 10 N hydrochloric acid (Thermo Fisher Scientific, USA) in a small Erlenmeyer flask, and the final volume was adjusted to 100 ml by adding 98 ml of distilled water. The flask was covered with aluminum foil to protect it from light exposure. A magnetic stirrer was placed inside the flask, and the mixture was stirred on a shaker for 30 min at room temperature until fully dissolved. Next, 1 ml of the tissue homogenate supernatant from each sample was mixed with 2 ml of the aforementioned solution (the tube lids were covered with foil or parafilm) and incubated in a boiling water bath for 45 min. After this period, the tubes were cooled to room temperature and centrifuged at 1500 rpm for 10 min. The absorbance of the upper layer of the solution in each tube was measured at a wavelength of 535 nm using a spectrophotometer (Thermo Fisher Scientific, Waltham, MA, USA), and the MDA concentration was calculated using the formula below. The results were reported in units per gram of tissue. In this formula, C represents the MDA concentration and A represents the measured absorbance:

$$C (M) = A / 1.65 \times 105 \quad (1)$$

1.3.2. Measurement of Total Thiol Groups

In this study, the DTNB reagent (Abcam, Cambridge, UK) was utilized to measure the total thiol groups. This reagent reacts with sulfhydryl groups (SH) to produce a yellow compound that exhibits an absorbance peak at 412 nm. For this assay, 50 µl of the tissue homogenate sample was mixed with 1 ml of Tris-EDTA buffer (30 mM Tris, 3 mM EDTA, pH 8.2; Sigma-Aldrich), and the absorbance was recorded at 412 nm

against the Tris-EDTA buffer alone (A1). Subsequently, 20 µl of DTNB reagent (10 mM in methanol) was added, and after 15 min at room temperature and in the dark, the absorbance of the samples was measured again (A2). The absorbance of the DTNB solution alone was also recorded as a blank (B). Finally, the total thiol (SH) group concentration (mM) was calculated using the following formula. Results were reported in units/g of tissue:

$$\text{SH concentration (mM)} = \frac{(A2 - A1 - B) \times 1.07}{(0.05 \times 13.6)} \quad (2)$$

1.3.3. Measurement of Superoxide Dismutase (SOD) Activity

The activity of the SOD enzyme was measured using the colorimetric method. This method is based on the production of superoxide through the auto-oxidation of pyrogallol and the inhibition of MTT reduction by superoxide to form insoluble formazan crystals. The reaction is halted by the addition of dimethyl sulfoxide (DMSO), which dissolves the formed formazan and stabilizes the resulting color. In brief, 60 µl of the homogenized tissue, MTT (Thermo Fisher Scientific), and pyrogallol (Sigma-Aldrich) were added to the wells and incubated for 5 min at room temperature in the dark. The reaction was stopped by adding DMSO (Merck), and the absorbance was measured at 570 nm using an ELISA reader (BioTek Instruments, Winooski, VT, USA). The results were expressed in units/g of tissue.

1.3.4. Measurement of Acetylcholinesterase (AChE) Activity

In this study, the Ellman method was employed to measure AChE activity. Specifically, 0.217 g of acetylthiocholine iodide (Sigma-Aldrich) was dissolved in 10 ml of distilled water. This solution was aliquoted into 0.4 ml volumes in Eppendorf vials and stored frozen at -20 °C. Subsequently, 3 ml of phosphate buffer (Merck), along with 40 µl of the tissue homogenate supernatant and 100 µl of DTNB solution (Abcam), were mixed in a test tube. Using a micropipette, 10 µl of thiocholine was added to the tube. The initial absorbance at 412 nm was recorded immediately after the addition of thiocholine. After 10 min, the secondary absorbance was measured, and enzyme activity was calculated using the formula below:

$$R = 5.74 (10^{-4}) \left[\frac{\Delta A}{C_0} \right] \quad (3)$$

- R = moles of hydrolyzed substance per g of tissue per min
- ΔA = change in absorbance per min
- C₀ = initial tissue concentration

1.4. Cell Counting Method in Histological Studies

Initially, stained slides were examined using a light microscope equipped with a computer-connected camera (Olympus BX51, Japan) at a magnification of 40x. Images of the hippocampal regions were captured and transferred to a computer for analysis. Dark neurons, as well as NF-kB positive and DCX positive cells, were counted separately in the corresponding sections using ImageJ software (National Institutes of Health, Bethesda, MD, USA). The cell density was calculated using the following formula:

$$N_A = \frac{\Sigma Q}{\frac{a}{f} \times \Sigma P} \quad (4)$$

In this formula, N_A represents the numerical density or the number of cells per unit area. ΣQ denotes the total number of counted cells, ΣP indicates the number of counted frames, and a/f refers to the area of each frame, measured in mm^2 .

1.3. Histochemical of Spinal Cord

At the end of a four-week experimental period, all animal subjects across the various groups were euthanized under deep anesthesia. Following euthanasia, the animals were perfused with a 10% formalin solution ($n=7$). The spinal cords were extracted, and paraffin blocks were prepared, ensuring that the lesion area was included (+1 mm from the lesion center). Tissue sections were subsequently obtained using a Rotary microtome (MR 3000, Histo-Line, Italy), yielding slices with a thickness of 5 μm and an inter-slice distance of 25 μm (H).

For the purpose of conducting CV/LFB staining, the paraffin-embedded spinal cord sections were initially deparaffinized and rehydrated. The tissue was then subjected to LFB (Sigma-Aldrich, St. Louis, MO, USA) staining for 18 hours at 60°C, followed by a CV (Merck) staining process lasting 10 minutes at 40°C. Once staining was complete, the samples were mounted for observation. The dorsal myelin regions were examined under a light microscope equipped with a camera (BX51, Olympus, Japan), and images were captured using a 40x objective lens. Hematoxylin and eosin (H&E; Thermo Fisher Scientific, Waltham, MA, USA) staining was employed to analyze the presence and distribution of inflammatory cells within the dorsal horn region. This technique facilitated the visualization of cellular morphology and allowed for a detailed examination of the inflammatory response in the tissue samples.

2. Results

2.1. Compensatory Movement Processes in the Hindlimbs of Spinal Cord-Injured Animals Treated with a combination Scaffold

The results regarding motor deficits caused by spinal cord injury and the impact of scaffold implantation treatment on the improvement of motor parameters, as assessed by the BBB test, are presented in Figure

1S. On the first day post-surgery, all animals recorded average BBB scores of less than 2, confirming the successful induction of dorsal hemisection of the spinal cord. During the first week following the procedure, no significant changes were observed in the recovery of hindlimb movement among the animals. However, in the second week, both the PCH and PCHN groups demonstrated a notable increase in motor improvement compared to the control group ($p<0.01$ and $p<0.0001$, respectively). Additionally, the PCHN group exhibited significant improvement relative to the PCN group ($p<0.01$). The differences among the groups in the third week were similar to those observed in the second week. By the fourth week, all groups that received scaffold implantation showed significant enhancements in hindlimb movement compared to the control group ($p<0.001$ for PCN and $p<0.0001$ for the other groups). Furthermore, the motor deficits in the PCN group were significantly greater than those in the PCNH and PCH groups ($p<0.0001$ and $p<0.05$, respectively). These findings indicate that the implantation of PU/f-MWCNT/Hsd@lip, with or without cells, leads to improvements in motor deficits in the hind limbs of animals after four weeks.

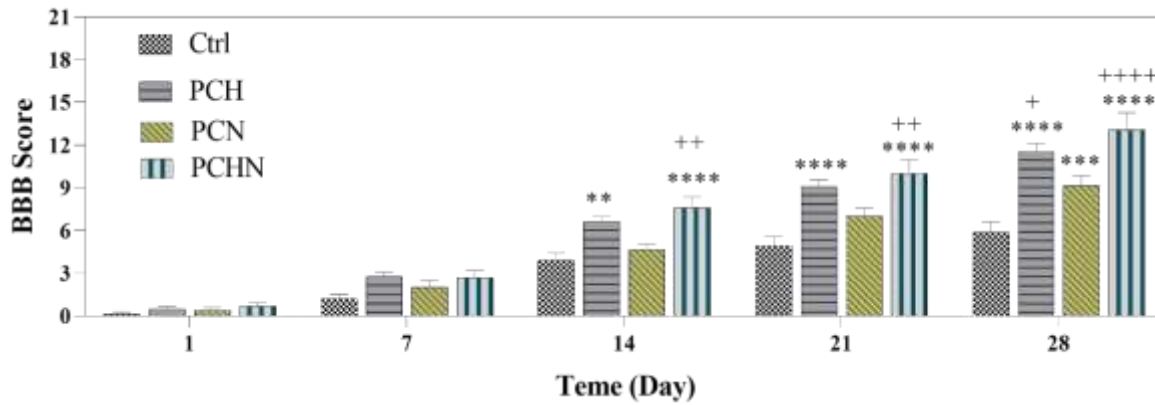


Figure 1S. Results of BBB behavioral scale of animals in the control, PCH, PCN and PCHN groups on days 1, 7, 14, 21 and 28 after spinal cord injury. Data are shown as mean and SEM ($n=15$). Asterisk indicates a significant difference between groups and the control group at each time point. + indicates a difference compared to the PCN group. PCN indicates spinal cord injured animals containing PU/f-MWCNT/NSCs scaffolds, PCH indicates spinal cord injured animals receiving PU/f-MWCNT/Hsd@lip scaffold and PCHN indicates spinal cord injured animals receiving PU/f-MWCNT/Hsd@lip/NSCs scaffolds

2.2. The effect of combination scaffolds on the histochemical structure of spinal cord tissue

Figure 2S presents histochemical images of spinal cord tissue sections at the lesion site following the fourth week of the study. The histochemical analysis reveals that the control group experienced significant structural disruption due to microglial activation. In contrast, the histological observations in the PCN group

indicated that the scaffolds containing neural stem cells lacked sufficient efficacy to modulate inflammation in the dorsal horn region. However, the histological results for the PCH and PCHN groups demonstrated that the implantation of these scaffolds into the injured spinal cord enhanced tissue structure by reducing activated microglial activity and stabilizing the neural architecture in this segment.

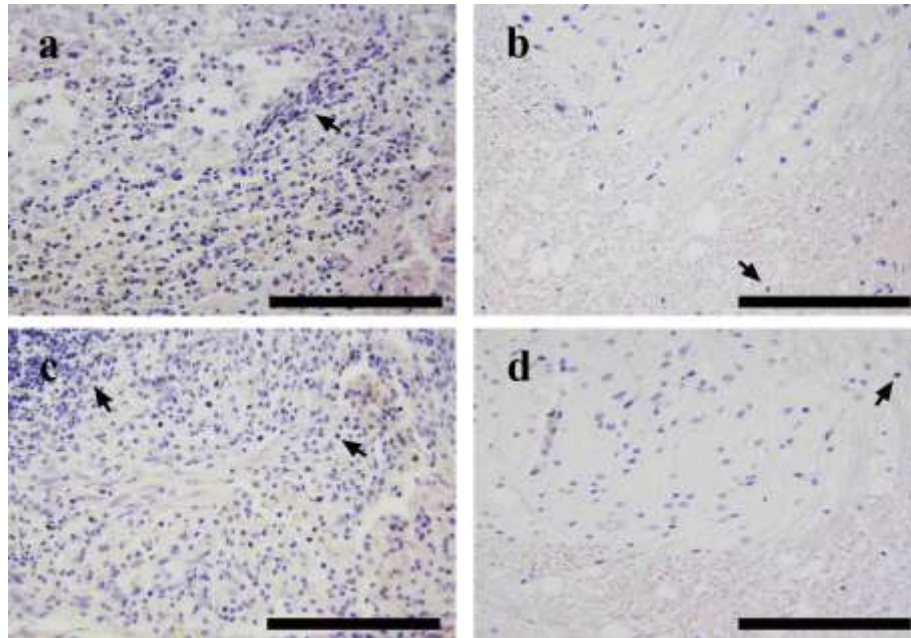


Figure 2S. The micrographs display histochemical images of tissue sections from animals four weeks post-spinal cord injury, utilizing Luxal Fast Blue/Cresyl Violet staining. The black arrow highlights several microglial cells in the dorsal horn region. The images were captured at 40x magnification, and the scale bar represents 100 microns. PCN refers to spinal cord-injured animals treated with PU/f-MWCNT/NSCs scaffolds. PCH indicates spinal cord-injured animals that received PU/f-MWCNT/Hsd@lip scaffold implantation. PCHN denotes spinal cord-injured animals treated with PU/f-MWCNT/Hsd@lip/NSCs scaffolds.

Figure 3S illustrates the outcomes of Luxal Fast Blue staining in the dorsal corticospinal tract following the intervention and induction of dorsal hemisection in rats. This tract is crucial as it transmits movement signals from the cerebral cortex to the limbs. The findings indicate significant myelin and tract destruction in this region, resulting in tissue loss, which is a consequence of spinal cord injury in the control group. Treatment with tissue-engineered scaffolds showed a substantial restoration of myelin after the intervention, yielding promising results in histological evaluations.

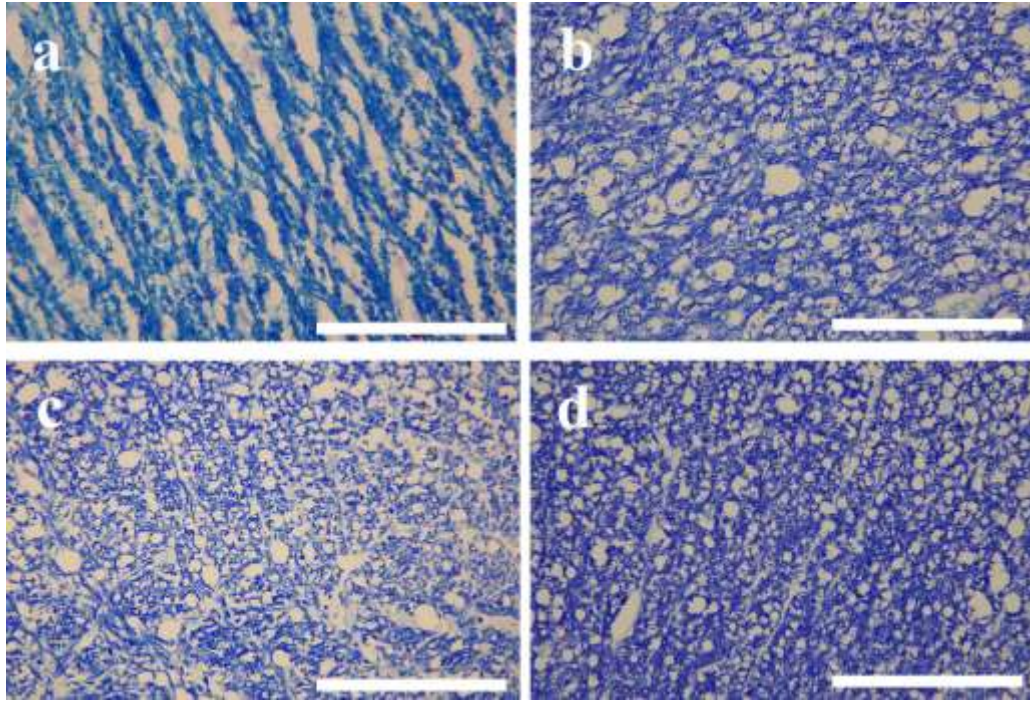


Figure 3S. The micrographs display histochemical images of tissue sections from animals four weeks post-spinal cord injury, utilizing Luxal Fast Blue staining. The images were captured at 40x magnification, and the scale bar represents 100 microns. PCN refers to spinal cord-injured animals treated with PU/f-MWCNT/NSCs scaffolds. PCH indicates spinal cord-injured animals that received PU/f-MWCNT/Hsd@lip scaffold implantation. PCHN denotes spinal cord-injured animals treated with PU/f-MWCNT/Hsd@lip/NSCs scaffolds.

2.3. Correlation of changes in electrophysiological recordings with measured parameters in the hippocampus following spinal cord injury

To analyze the changes in the action potential of hippocampal neurons, we conducted a linear regression of electrophysiological recordings, focusing on the percentage change in amplitude from baseline in relation to parameters such as AchE, DCX, MDA, NF-KB, avoidance memory (PA), -SH, SOD, and dark neurons. The findings indicated no significant correlation between long-term potentiation (LTP) and DCX ($R^2 = 0.0003$, $p = 0.92$). However, there were significant relationships with several parameters: MDA ($R^2 = 0.38$, $p = 0.0003$), -SH ($R^2 = 0.17$, $p = 0.02$), PA ($R^2 = 0.18$, $p = 0.02$), dark neurons ($R^2 = 0.3$, $P < 0.0001$), DCX ($R^2 = 0.29$, $p = 0.002$), and NF-KB ($R^2 = 0.19$, $p = 0.017$). Nevertheless, the coefficient of determination (R^2) for all these parameters was below 40%, indicating a low reliability of their relationships and suggesting a lack of correlation with synaptic dysfunction in the hippocampus. In contrast, the analysis of the relationship between electrophysiological changes and AChE levels in the hippocampus revealed a significant correlation ($p < 0.0001$), with an estimated R^2 of 0.73. This suggests that 73% of the variability

in AchE can be explained by changes in LTP. Consequently, following spinal cord injury, there is a notable positive relationship between the increase in action potential dysfunction of hippocampal neurons and the rise in hippocampal AchE levels.

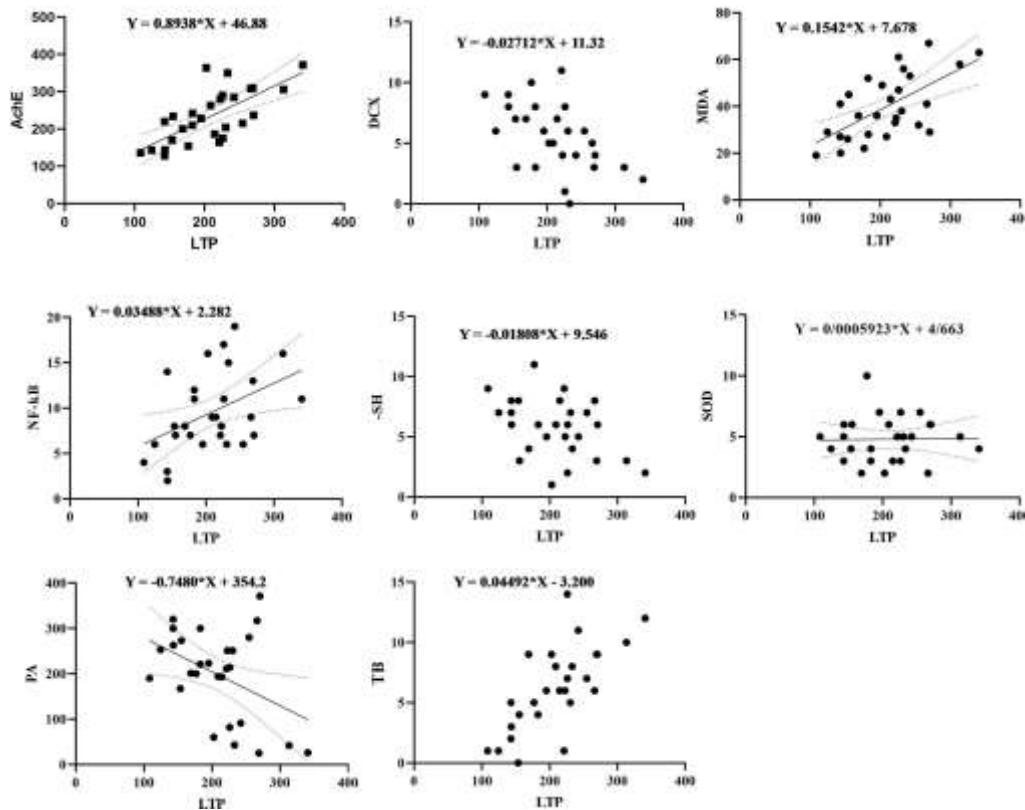


Figure 4S. linear regression of electrophysiological recordings, focusing on the percentage change in amplitude from baseline in relation to parameters such as AchE, DCX, MDA, NF-KB, avoidance memory (PA), -SH, SOD, and dark neurons following spinal cord injury.